

Suppressed but Unmoved:

Defensive Suppression and Failed Offensive Adaptation in the NBA

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Introduction. A coach watches a play that was working suddenly stop working. Execution? Random variation? Or has the defense taken it away? Coaches talk constantly about adjustments, yet NBA offenses rarely make them. More than nine times in ten, teams keep calling the same play at nearly the same rate after it breaks down. This paper argues that in many cases the defense has already won, and continuing the play is a costly mistake. This matters for the industry because if such collapses are real and go unaddressed, teams are forfeiting points in real time using information that is already sitting in front of them mid-game.

Methods. The analysis draws on 863,752 classified NBA possessions from the 2022–23 through 2025–26 seasons (5,234 games), built from public play-by-play through the NBA Stats API, with a calibrated play-type classifier covering pick-and-roll, spot-up, drive, post-up, cut, pull-up, transition, and putback. A play type is flagged as a suppression trigger when a team's first-half points per possession exceeds its second-half figure by 0.25 or more, with at least five possessions in each half. Because triggers are selected on an elevated first half, a raw before-and-after comparison conflates suppression with ordinary mean reversion. This is corrected with a same-game matched-pair design: each triggered play type is paired with the opposing team running the same play type in the same game, matched on first-half efficiency, with cluster-robust and permutation inference, plus caliper, blowout, and overtime-boundary robustness checks.

Results. Defensive suppression holds up net of regression to the mean, with matched pairs showing an excess decline of 0.408 points per possession (95% CI 0.378 to 0.438, $N = 1,136$ pairs across 848 games, $p < 0.0002$), stable across every robustness check. Offenses largely fail to adapt: teams continue running the suppressed play in 93% of events, at 97 to 98% of prior volume, at a cost of roughly 5.4 to 5.5 points per trigger. Substituting the primary ball-handler does not recover efficiency (a negligible 0.039 point difference). Even when a higher-scoring alternative already existed in the same game, true of 95.3% of triggers, no team systematically reallocated toward it. Matchup-ranked counters, validated out of sample, outperform continuation by 0.445 points per possession in the playoffs and 0.584 in the regular season, while team-level suppression severity and unrealized counter value show only a weak relationship with win rate ($r = +0.25$).

Conclusion. Defensive suppression is a genuine, sizable, and largely unaddressed source of offensive inefficiency. Teams continue running actions the defense has already solved, even when better options are available in the same game, and a simple first-half signal can identify both the collapse and its remedy. This suggests a practical application for the industry: real-time tools that flag suppression as it happens and rank live counters. Whether teams acting on the signal can convert it into realized efficiency remains an open question for future work.